ASTERIX PVT LTD

Guardians of the Wild, Engineered.

"Ground when possible. Air when necessary. Disturb nature as little as possible."

Response to E-Tender EGNI/2026/AEOUV-04

Autonomous Electric Off-road Utility Vehicle for Rainforest Harmony National Park

Priyadharshan S	Member 2	Member 3	Member 4
CEO	CTO	CFO	COO

PROPOSAL HOOK

Executive Summary: Pitch Overview

company specializing in zero-impact conservation robotics platforms and mesh sensor arrays.

- The Ask: ₹3.2 Cr grant funding to deliver a turnkey AEOUV fleet + Smart-Pod communications mesh network.
- The Offer: A Ground-Air Swarm integration combining autonomous electric vehicles (AEOUV), biomimetic drones, and passive forest sensor nodes.
- Capability Focus: Zero-disturbance wilderness monitoring, automatic seed drop reforestation,

Swarm Integration Offering

A cooperative system where the ground base station supplies energy & calculations, and drones provide reach.

PERFORMANCE TARGET

Executive Summary: Key Performance Indicators

of thick jungle canopy per system daily via coordinated aerial patrols.

- **Eco Footprint:** Zero tailpipe emissions. Eliminates the use of destructive diesel off-road vehicles.
- **Emergency Response:** Drone dispatch locates anomalies and drops medical aids in under 20 minutes.
- **Reliability Target:** Guaranteed 95% autonomous uptime with self-healing routing

120 km²

DAILY SWARM COVERAGE

< 20 min

EMERGENCY RESPONSE TIME

95%

OPERATIONAL ENVIRONMENT

Rainforest Harmony: Environmental Constraints

- **Vast Territory:** 3,885 km² of equatorial rainforest containing rich biological species and sensitive ecosystems.
- **Extreme Weather:** Temperatures of 25–35 °C with high humidity and heavy rainfall (~3,048 mm/yr). Wet season runs Nov–May.
- **Terrain Hurdles:** Thick clay mud, seasonal flooding, and dense overhead canopy that blocks cellular networks.

Harmony National Park Metrics

Average Temperature: 25–35 °C
Annual Precipitation: ~3,048 mm
Wet Season: Nov–May

SCOPE OF WORK

The Mission Mandate: Four Roles

- **1. Wildlife Monitoring:** Continuous, non-intrusive tracking of species migration patterns, population health, and poacher movements.
- **2. Forest Management:** Mapping foliage canopy density, checking tree health, and carrying out aerial seeding.
- **3. Medical Administration:** Automated, remote target delivery of dosed vaccines to wild animals.
- **4. Zero Disturbance:** Maintaining zero habitat footprint, avoiding soil compaction, and using silent mechanics.

Wildlife Monitoring **Forest Seeding**
Remote Vaccination **Acoustic Silencing**

RFP requires strict compliance with zero-footprint operation parameters.

PROBLEM STATEMENT

The Territory Gap: Why Traditional Methods Fail

beats are exposed to wild animal attacks, insect-borne diseases, and flash floods.

- **Heavy Vehicle Damage:** Conventional diesel utility vehicles compact forest soils, create deep ruts, and damage root structures.
- **Acoustic Disturbance:** Loud engines scare nesting birds and disrupt animal migration pathways.
- **RFP Conclusion:** Government tender notes that "traditional methods prove insufficient" for

Traditional Patrol Bottlenecks

• Heavy Soil Compaction
• Engine Noise (>85 dB)
• Communication Blindspots

CORPORATE PROFILE

Company Overview & Chennai Facility

- OEM Location: Headquartered with primary assembly plant in **Chennai, Tamil Nadu**.
- Incorporation Status: Fully registered, Indian-owned private limited company specializing in environmental robotics.
- Workforce Size: 42 full-time staff, incorporating a **40/60 diversity split** across core engineering and research teams.
- Focus: Designing clean, silent ground platforms, autonomous payload bays, and low-power mesh radios.

Chennai Assembly Facility

- Subsystem Fabrication Line
- Sensor Calibration Chamber
- Outdoor Canopy Sandbox Testbed

ORGANIZATIONAL STRUCTURE

Leadership Team & Organizational Design

- **Priyadharshan S (CEO):** Directs corporate strategy and anchors government relations.
- **Member 2 (CTO):** Directs edge AI, autonomy navigation stacks, and sensor fusion.
- **Member 3 (CFO):** Manages unit economics, BOM optimization, and capital allocation.
- **Member 4 (COO):** Oversees manufacturing, supply chain scaling, and WPC/DGCA filings.

Board Mandate

Our lean, matrix organization coordinates mechanical design, software validation, and environmental research under one corporate loop.

PRODUCT OFFERING

System-of-Systems: 5-Layer Stack

- 5. Command & Safety: HQ station controls teleoperation and handles health diagnostic overrides.
- 4. Aerial Swarm: VTOL bird-drones carrying modular pods to hard-to-reach zones.
- 3. Actuation Layer: Onboard 5-DOF robotic arms and Adaptive Mission Bay swapper.
- 2. Mobility Platform: Electric BAJA off-road vehicle with range-extending brakes.
- 1. Sensing & Mesh: Solid-state LiDAR scanners + tree-mounted Smart Pod network.

5. Command & Safety Layer

4. Aerial Layer (EcoWing-U Drones)

3. Actuation Layer (Arms + AMB)

2. Mobility (AEOUV Electric ATV)

1. Sensing & Mesh Network

GROUND PLATFORM SPECS

AEOUV Buggy: Mechanical & Chassis Base

- **Chassis structure:** High-strength tubular steel space frame (compliant with BAJA off-road standards).
- **Power & Motor:** 9 kW Brushless DC (BLDC) motor with regenerative braking.
- **Battery:** AIS-156 Certified Lithium-Ion Pack (80 Ah Capacity, 72V nominal).
- **Suspension:** Double-wishbone suspension with 8-inch travel for muddy trails.

Mechanical Parameter	Specification
Suspension Travel	8 Inches (Double-wishbone)
Chassis frame	Tubular steel space frame
Powertrain	9 kW BLDC (AIS-156 compliant)
Braking	Regenerative range extension

AUTONOMY SUITE

AEOUV Buggy: Sensor Fusion & Autonomy Stack

- **Autonomy Compute:** NVIDIA Jetson Orin Core running edge computer vision.
- **Sensor Stack:** Rotating solid-state LiDAR, stereo depth cameras, mmWave radar, and RTK-GNSS.
- **Path Planning:** Local costmap generation that automatically marks forest quiet zones as no-go polygons.

Edge Navigation Core

Autonomy stack allows the buggy to swerve around fallen trees and navigate deep forest paths without active internet.

ACTUATION SYSTEMS

AEOUV Buggy: Dual 5-DOF Robotic Manipulators

- **Twin Manipulators:** Two 5-DOF robotic arms mounted on the chassis.
- **Water Siphon:** Reaches into riverbeds to siphon water samples, running chemical diagnostics.
- **Soil Boring:** Bores soil samples along path margins.
- **Fail-Safe:** Actuators release in under 100 ms during manual driver overrides.

Water & Soil Sampling

Arm draws samples and passes them to internal diagnostic chambers, eliminating manual exposure.

AERIAL PLATFORM SPECS

EcoWing-U Drone: Hybrid VTOL Aerodynamics

- **Hybrid Propulsion:** Twin vertical lift rotors paired with a rear pusher propeller for high-speed forward glide cruise.
- **Avionics Platform:** PX4 Flight Controller paired with visual odometry (VIO) for GPS-denied canopy flight.
- **Airframe structure:** Lightweight, bio-mimetic carbon fiber shell modeled after forest birds.

Bio-Mimetic UAV Core

Wingspan: 1.2 meters
Weight: 3.0 kg
Glide Range: 25 km

Adaptive Mission Bay (AMB): Swapping Loop

- **Weatherproof Chamber:** Enclosed cargo bay housing a magazine of six hot-swappable Mission Pods.
- **Robotic Linear Shuttle:** Translates, secures, and magnetically latches pods onto the drone landing carriage.
- **Electro-Permanent Magnets:** Delivers rigid structural locking without active power draw.
- **Swap Speed:** Completes payload swaps in under 30 seconds.

EPM Magnetic Docking

Automatic pogo-pin contacts link power and data lines instantly during pod swapping.

Modular Mission Pods: Vaccination & Seed Drop

Vaccination Pod

- CO_2-driven micro-dart launcher.
- Targeting camera array tracks animals.
- Delivers vaccines with precision.

Seed Planting Pod

- Pressurized seed hopper.
- Drops biodegradable capsule seeds.
- Slow-release fertilizer shell.

MODULAR PAYLOADS

Modular Mission Pods: Sampling & Emergency

Environmental Sampling Pod

- Retractable air siphon tube.
- Water container with preservative chemistry.
- Onboard diagnostics reads basic pollutants.

Emergency Delivery Pod

- Fitted with first-aid kits and anti-venoms.
- Includes satellite signaling beacons.
- Mechanical drop release mechanism.

SENSING NODES

Smart Forest Pod Network: Passive Node Sensors

- **Continuous Auditing:** Low-power nodes placed in trees and riverbeds by drones.
- **Sensor Breakdown:**
  • *Acoustic*: Identifies chainsaws, gunshots, and engines.
  • *Hydrological*: Measures water temperature, pH, and pollutants.
  • *Camera*: Motion-activated visual logs.
- **Solar Charging:** Integrated solar panels keep batteries functioning over 12 months.

Persistent Sensing Nodes

Smart Pods provide passive 24/7 monitoring, waking up the network only when anomalies are detected.

NETWORK DESIGN

Mesh Communication: Self-Healing Network

- **Self-Healing Protocols:** Sub-GHz LoRa mesh routes signals from pod to pod around obstacles.
- **Resilience:** Operates locally where cellular and public networks fail.
- **Licensing:** Requires WPC (Wireless Planning & Coordination) spectrum clearance for Indian forest zones.

LoRa Mesh Topology

If Pod A is blocked, data automatically hops via Pod B and C to reach the nearest AEOUV ground hub.

MISSION SCENARIO I

Concept of Operations: Morning & Midday

06:00	09:00	12:00
Acoustic Pod #2 detects chainsaw. Mesh alerts AEOUV.	AEOUV deploys Drone 1 with Seed Pod to steep slope.	Drone 2 launches with Vaccination Pod to treat target leopard.

MISSION SCENARIO II

Concept of Operations: Evening & Night

15:00
Robotic arm siphons river water; runs local diagnostics.

18:00
Buggy A suffers battery fault; sister Buggy B rescues.

21:00
Lost trekker detected via thermal; aid dropped by drone.

TENDER ALIGNMENT

Compliance Matrix: RFP Requirements vs. ASTERIX

RFP Mandatory Requirement	ASTERIX Proposed Feature	Compliance Status
Wildlife Monitoring	Passive sensor pods + drone thermal classification.	Fully Compliant
Reforestation	Robotic arm seed borer + drone capsule drops.	Fully Compliant
Medical Administration	Pressurized Vaccination Pod with target tracking.	Fully Compliant
Minimal Disturbance	Decibel-damped bird-drones + trail-bound pathing.	Fully Compliant

COMPETITIVE EDGE

ASTERIX Unique Selling Points (USPs)

- **1. Silent Bio-Mimetic UAV:** Avoids wildlife startle responses typical of commercial multirotors.
- **2. Buggy-Rescues-Buggy Autonomy:** Integrated V2V recovery script ensures vehicle redundancy in deep ranges.
- **3. Integrated Human SAR:** Real-time search-and-rescue via thermal drone tracking and emergency pod deployment.
- **4. On-Field diagnostics:** Integrated physical sampling arms + testing lab inside pods.

Silent UAV **V2V Recovery** **Human SAR** **On-Field Diagnostics**

Our edge centers on unified coordination rather than standalone vehicles.

ENGINEERING RISK

Technical Feasibility & Make-vs-Buy Stance

- **Integration Risk Stance:** Focusing custom engineering strictly on proprietary IP (AMB payload swapper, software coordinate AI).
- **Buy Category:** LiDAR sensors, motors, automotive compute modules, battery cells.
- **Make Category:** Custom carbon fiber drone structure, swapper mechanics, mesh node electronics.

BUY (PROVEN BUILDING BLOCKS)

LiDAR sensors, compute boards, battery packs.

MAKE (PROPRIETARY IP)

Adaptive swapper, LoRa mesh nodes, coordinator AI.

QUALITY ASSURANCE

Prototype Validation: Sandbox Forest Pilot

- **Sandbox Phase:** Indoor subsystem validation followed by sandbox testing in a dense forest plot in Tamil Nadu.
- **Supervised Pilot:** Deployment of a 2-AEOUV + 10-pod network in Harmony National Park.
- **Validation Goals:** Confirming drone VTOL stability under wind, mesh packet transmission rates, and autonomous buggy path safety.

TRL Development Path

Lab Proof (TRL 4) ➔ Forest Sandbox (TRL 5-6) ➔ National Park Supervised Pilot (TRL 7).

TIMELINE

Prototype Validation: Timeline & Gantt

- Month 1-3 (TRL 4): Frame assembly, sensor integration, and indoor arm calibration.
- Month 4-7 (TRL 5-6): Outdoor sandbox testing, mesh range validation, and pod swapping debug.
- Month 8-12 (TRL 7): Supervised forest park pilot. Initial 3-month driver-in-loop tracking.

Validation Milestones

• **M3**: Integrated chassis & computing check.
• **M7**: 5 km mesh communication range test.
• **M12**: Handover of pilot data to forest department.

OPPORTUNITY

Market Analysis: TAM-SAM-SOM

- **TAM (India):** ₹150 Cr, based on ~100+ national parks and 500+ sanctuaries seeking modern monitoring gear.
- **SAM:** ₹45 Cr, targeting parks with dense foliage and high budgets.
- **SOM (Beachhead):** ₹12 Cr, focusing on protected areas in Northeast India.
- **Assumed Spend:** ₹3 L/km² annual conservation technology spending.

TAM-SAM-SOM Funnel

TAM: ₹150 Cr → SAM: ₹45 Cr → SOM: ₹12 Cr.

STRATEGIC ANALYSIS

PESTEL & SWOT Strategic Assessments

SWOT HIGHLIGHTS

- **S**: Modular AMB swapper, silent drone IP.
- **W**: High initial drone R&D cost.
- **O**: Verifiable carbon offset audits.
- **T**: Legal clearances (BVLOS).

PESTEL HIGHLIGHTS

- **Political**: PLI Indian sourcing preference.
- **Environmental**: Zero soil rutting, zero noise.
- **Legal**: Complies with Wildlife Protection Act.

PRODUCTION LOGISTICS

Manufacturing Plan & Local Sourcing

- **Assembly Line:** Chennai assembly line handles chassis framework welding, wire harness routing, and sensor calibration.
- **Local Sourcing:** Complying with Indian PLI guidelines by sourcing chassis fabrication and batteries locally.
- **Low-Rate Production:** Targeting 3 prototype units for pilot phase, scaling to 10 systems/year.

Component	Sourcing Strategy
BLDC Motors	Indian Tier-1 Supplier (Buy)
Chassis Structure	Chennai Assembly (Make)
Fiber Shell Drones	Chennai Mold line (Make)

FIELD LOGISTICS

Service & Support: SLA & Depot Network

- **Local Depots:** Leveraging park warehouses as regional battery swap stations and spares depot hubs.
- **Uptime SLA:** 95% system availability guarantee under our standard AMC contract.
- **Spares SLA:** 24-hour shipping on critical parts from the Chennai central hub.
- **Desk Support:** 24/7 teleoperation desk handles sensor pings and overrides.

SLA Warranties

Our standard contract includes automatic "buggy-rescues-buggy" service swaps during mechanical faults.

FINANCIAL COSTING

Financial Model: Unit Economics

- **AEOUV BOM:** Frame & batteries (₹4.2 L) + Autonomy sensors (₹6.8 L) + Arms (₹2.4 L). Total BOM = ₹13.4 L.
- **UAV Unit Cost:** ₹3.8 L BOM (R&D-heavy low-acoustic composite airframe).
- **Smart Pod Cost:** ₹0.4 L BOM (low-power processor + solar cells).
- **Deployment Kit:** Total BOM ₹34 L. Sale Price: ₹48 L (Gross margin 41%).

Product Segment	BOM Cost	Selling Price	Margin %
AEOUV Hub	₹13.4 L	₹18.0 L	34%
EcoWing Drone	₹3.8 L	₹5.2 L	36%
Smart Pod	₹0.4 L	₹0.6 L	50%

FUNDING PROFILE

Financial Model: Capital & P&L Forecast

- **Capital Request:** ₹3.2 Cr, allocated to assembly tooling, prototype validation, and certification checks.
- **P&L Projections:**
  • Year 1: Hardware sales ₹48 L.
  • Year 3: Hardware sales ₹2.4 Cr + recurring service fees ₹80 L.
- **Recurring Revenue:** Conservation-as-a-Service (CaaS) tracking data packages drive long-term margins.

Use of Capital (₹3.2 Cr)

• Prototyping & R&D: 35%
• Plant Tooling: 20%
• Sandbox Testing: 15%
• Certification (DGCA/WPC): 10%
• Working Capital: 20%

INVESTMENT ANALYSIS

Investment Analysis: DCF & ROI Projections

- **Valuation framework:** DCF (12% discount factor) and market multiples (5x EBITDA) demonstrate corporate viability.
- **ROI Metrics:**
  • Modelled IRR: 24.2% over 5 years.
  • Net Present Value (NPV): Positive at ₹1.8 Cr.
  • Payback Period: 28 Months.

Modelled Profitability

Steady recurring software and telemetry service revenue yields healthy cash flow projections by Year 3.

RISK ASSESSMENT

Investment Analysis: Risk Register & Exit

- **Top Risks & Mitigations:**
  • *BVLOS drone rules*: Mitigate via early type certification files.
  • *Dense canopy navigation*: Mitigate via Visual Inertial Odometry (VIO).
- **Exit Strategy:** Strategic acquisition by defense/agri robotics firms, or technology licensing.

Risk Heatmap

• **BVLOS Approvals**: High risk / Mitigate via DGCA rules
• **Rainforest Humidity**: Med risk / Mitigate via IP67 chassis
• **Battery Drainage**: Med risk / Mitigate via solar cells

PROJECT ASK

Legal Stance, Roadmap & Ask

- Compliance Badges: Complies with DGCA rules, Bharatiya Vayuyan Adhinayam 2024, DPDP Act 2023, and Wildlife Protection Act 1972.
- Dispute Resolution: Headed by mutual mediation prior to arbitration under Chennai jurisdiction rules.
- The Ask: Requesting ₹3.2 Cr grant funding to deploy the ACS Swarm in Harmony National Park.

DGCA Compliant | DPDP 2023 | Wildlife Protection Act

"Engineering Conservation."